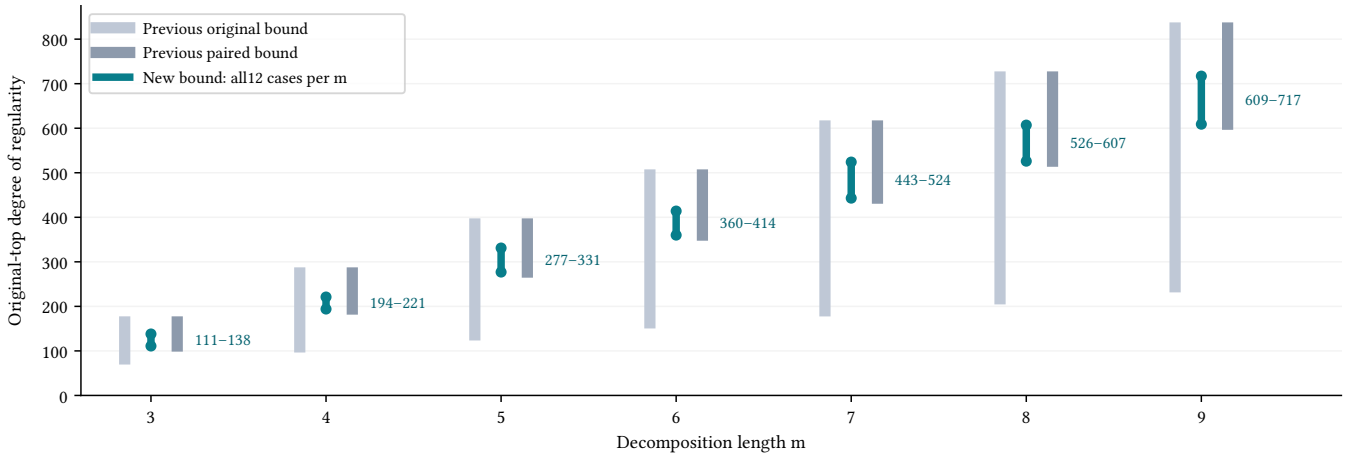
Stronger binary chain regularity bounds

84 saved inputs · 84 fresh replays · 84 S3-restored replays

All binary/Koblitz systems at m=3–9 passed. Each new interval holds for original, baseline and spike40 bases, planted and natural targets: 12 cases per m. These are tighter certificates on unchanged systems, not a measured base gain.

m	Previous original	Previous paired	New interval	Blocks
3	82–165	111–165	**111–138**	1
4	109–275	194–275	**194–221**	2
5	136–385	277–385	**277–331**	2
6	163–495	360–495	**360–414**	3
7	190–605	443–605	**443–524**	3
8	217–715	526–715	**526–607**	4
9	244–825	609–825	**609–717**	4



No exact full-chain degree is claimed. All intervals retain a gap. The bases still use 2^27 candidate x-coordinates per curve and remain fixed across m=2–9. Original planted targets differ from paired planted targets; repeated natural sources are provenance identities, not independent samples.

What this establishes. Nonadjacent chain components supply stronger upper bounds, while a checked coordinate projection strengthens the older original lower bounds. These identical overlapping intervals do not rank baseline and spike40. The prior exact m=2 degrees remain Koblitz 28, binary 28 and prime 42.

Certificate construction and preservation

Source binding. Check pinned source/input/base hashes; ring and equation IDs; every highest coefficient and full expected support; every variable-square field equation; and the exact saved witness suffix. Lower-degree coefficients remain unchanged by their source hash. No affine consequences are added.

Cubic components. Checked invertible coordinate transforms over the field of 2^83 elements give 27 disjoint cubic slots. Every coefficient of these relations is checked for both selector domains and both cubic shapes. Each slot permits at most two of its three variables; the component socle upper bound drops by 27.

Final quadratic component. Explicit original-equation combinations express all embedded products as scalar multiples of one product h for nonzero target. Variable squares imply every local variable annihilates h. Adding the missing product direction changes only degree 2. The full product quotient vanishes above degree 83, so the original quadratic component does too. Zero target is handled separately by invertible coefficient Frobenius. Small controls cover both branches.

Combine disjoint blocks. Select even-numbered S3 groups and verify that their variable blocks do not overlap. Tensor-product socle bounds add, giving `upper = N - 27*floor(m/2) + 1`, with `N = 27m + 83(m-2)`. Keeping only the first 27 selectors and all intermediate variables kills every non-field top monomial and gives `lower = 28 + 83(m-2)`.

Validation and scope. Five native tests pass, including 30 explicit small Macaulay/Hilbert cases and 128 independent arithmetic checks. Field irreducibility, matrices/inverses and all component relations are checked. Lean proves only slot-sum arithmetic, using standard propext and Quot.sound. The full ideal, annihilator, tensor and scalar-extension bridges are written/native evidence. Independent scientific review remains pending; these are producer replays.

Preserved data. All 360 raw manifest entries matched before and after S3-restored replay; all 84 immutable partial summaries remain. Source, inputs, full certificates, executable, logs and prior failures are retained. Data were committed and pushed before this report. Read `CHAIN_FORMAT.json`, `CHAIN_DESIGN.md`, `experiment-chain.export.json` and `evidence/chain-storage.json`.

Archive SHA256: `5da13e39bbc9db23f61a2fbd709da896bda04b1bb8eacbe78c218011758d579d`
Source: `8ccd4eb9729cb01767064e41c5507dcb0476b2ac`
Protocol: `ce0a47828dbf3f0dcfe311cefd43730ab671dcf9ac1da7e0cee8ecd2d9a1fb72`
Data commit: `19c6cc710c2acfafef5a20ad7847f48d44524428`

The content-addressed archive is in `s3://setec-astronomy-artifacts/store/sha256/` under its first two hash characters and full hash. Restore, check `manifest.sha256`, run `instrument replay ROOT`, and recheck the manifest.

Remaining work. Exact larger-m regularity/LFD, prime LFD and held-out elite-base studies. Higher-m prime top quotients are positive-dimensional in the saved presentation. Existing-host fallback followed retained EKS bucket denial. No timing, solving-degree, universal FFD/LFD or ECDLP attack claim.